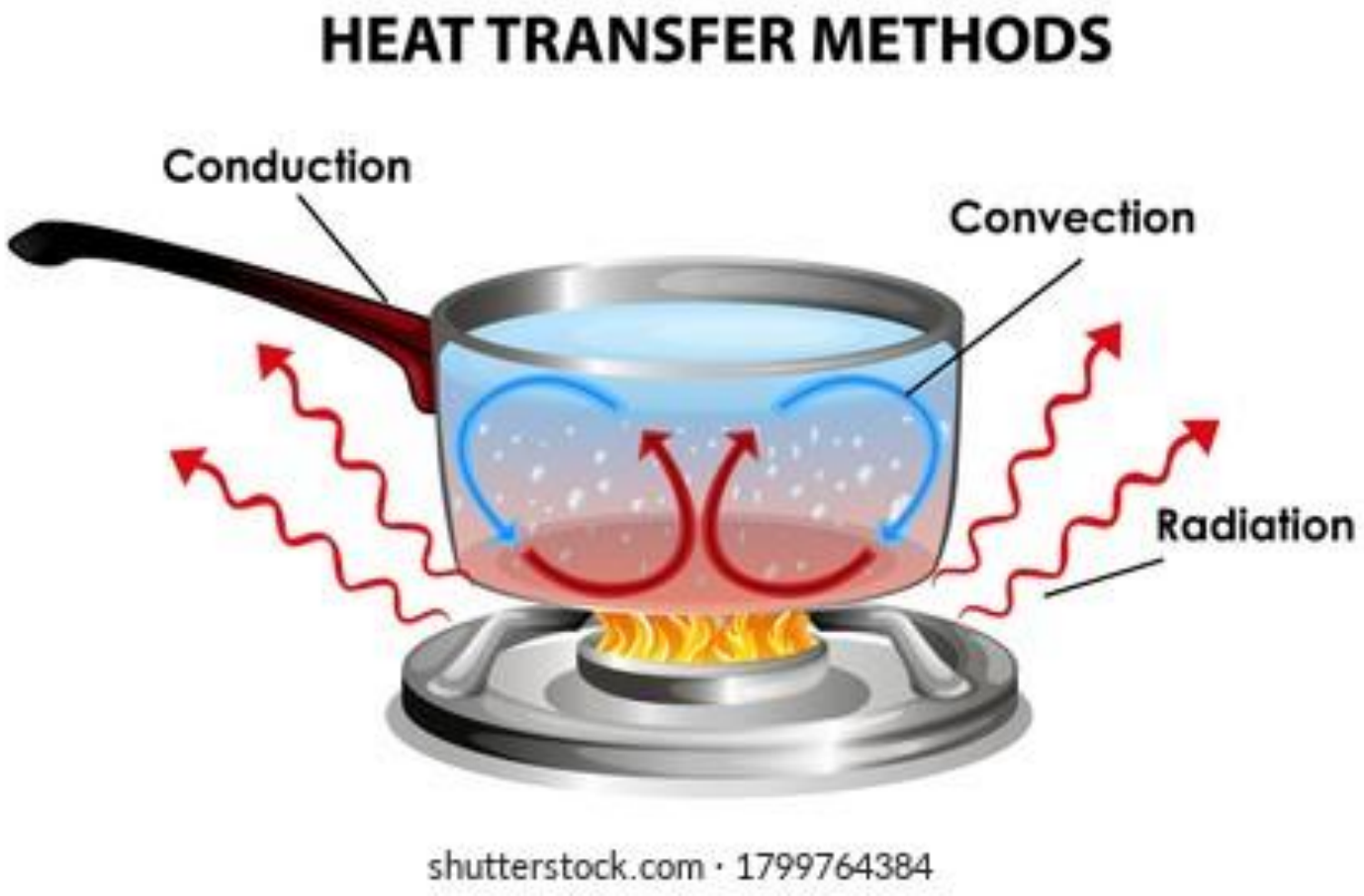
Modes of Heat Transfer

Heat transfer is the process by which thermal energy moves from one place to another. There are three primary modes of heat transfer:

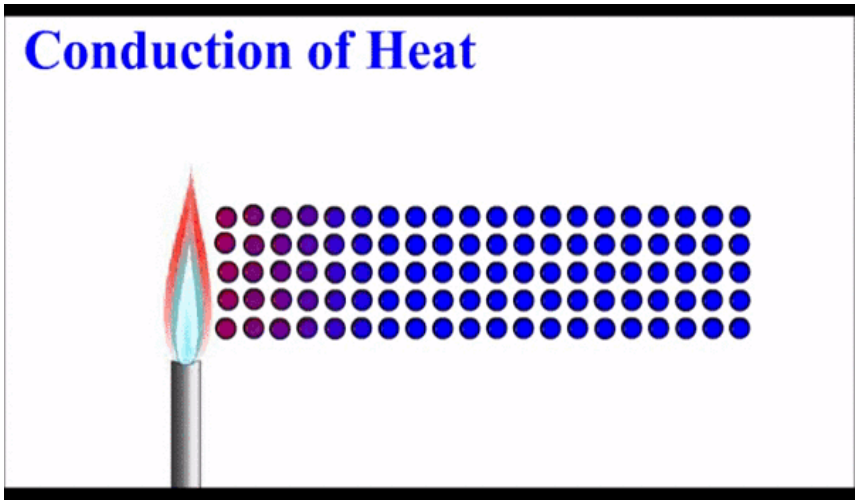


Conduction Modes of Heat Transfer

1. Conduction is the *transfer of heat through a solid material* from one molecule to the next. It occurs when molecules in a substance collide with each other, transferring energy.

How It Works: Heat moves from the hotter part of the object to the cooler part until the temperature is uniform.

Example: **Touching** a metal spoon that has been sitting in a hot pot – the heat travels from the hot end to the cooler end, eventually making the whole spoon hot.





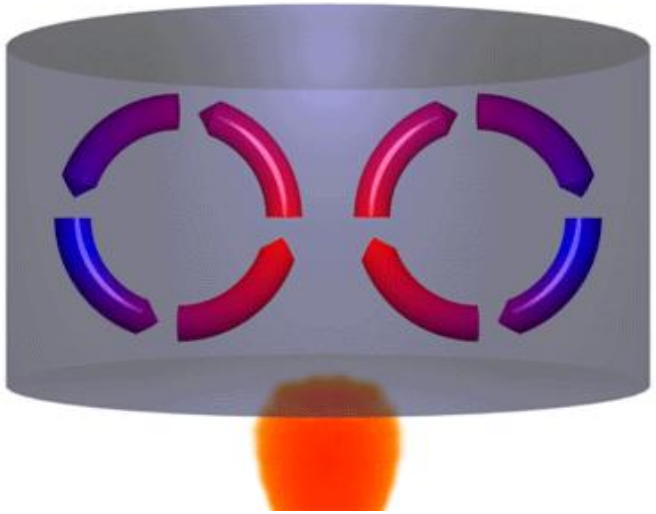
Convection Modes of Heat Transfer

2. Convection is the transfer of heat by the movement of fluids (liquids or gases). In convection, warmer areas of a liquid or gas rise to cooler areas, and cooler liquid or gas moves down to replace it, creating a circulation pattern.

How It Works: *Heat is carried by the fluid as it moves*, distributing thermal energy throughout the substance.

Example: Boiling water – the water at the bottom of the pot heats up, rises to the top, and cooler water moves down to take its place, creating a circular motion.

Real-World Application: Air Conditioning Systems: Cool air is circulated around a room, replacing warmer air and cooling the space.





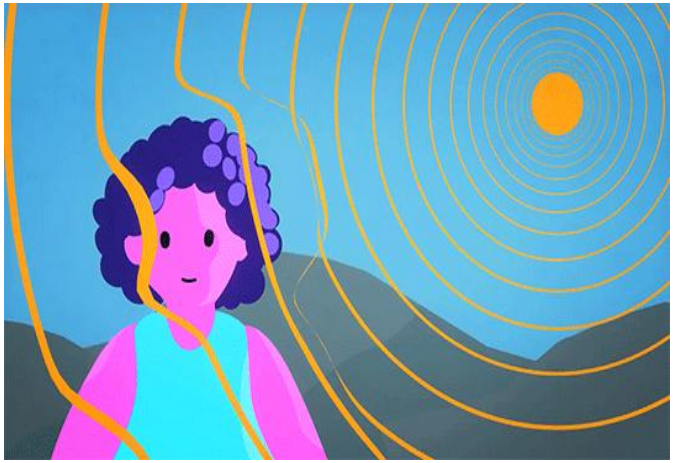
Radiation Modes of Heat Transfer

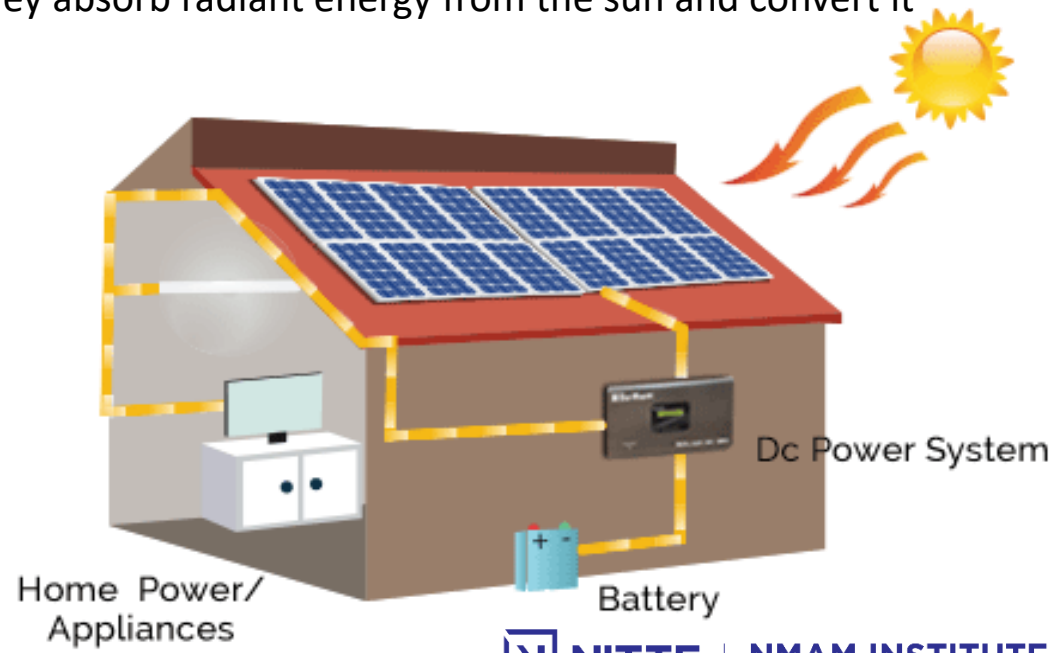
3. Radiation is the ***transfer of heat through electromagnetic waves***. Unlike conduction and convection, radiation does not require a medium (solid, liquid, or gas) to transfer heat; it can occur through a vacuum.

How It Works: *Heat energy is emitted as infrared radiation and can be absorbed by other objects.*

Example: Feeling the warmth of the sun on your skin – the heat travels through space and air by radiation.

Real-World Application: Solar Panels: They absorb radiant energy from the sun and convert it into electricity.



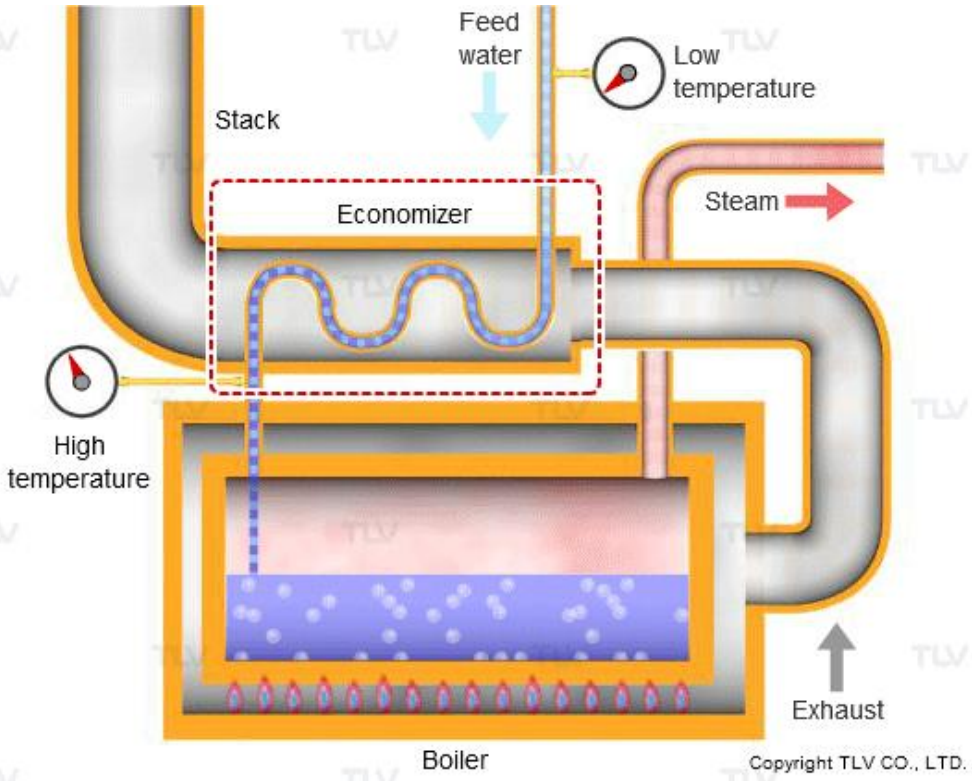


Summary of Heat Transfer Modes

1. Conduction: Heat transfer through direct contact in solids.

2. Convection: Heat transfer through fluid movement (liquids or gases).

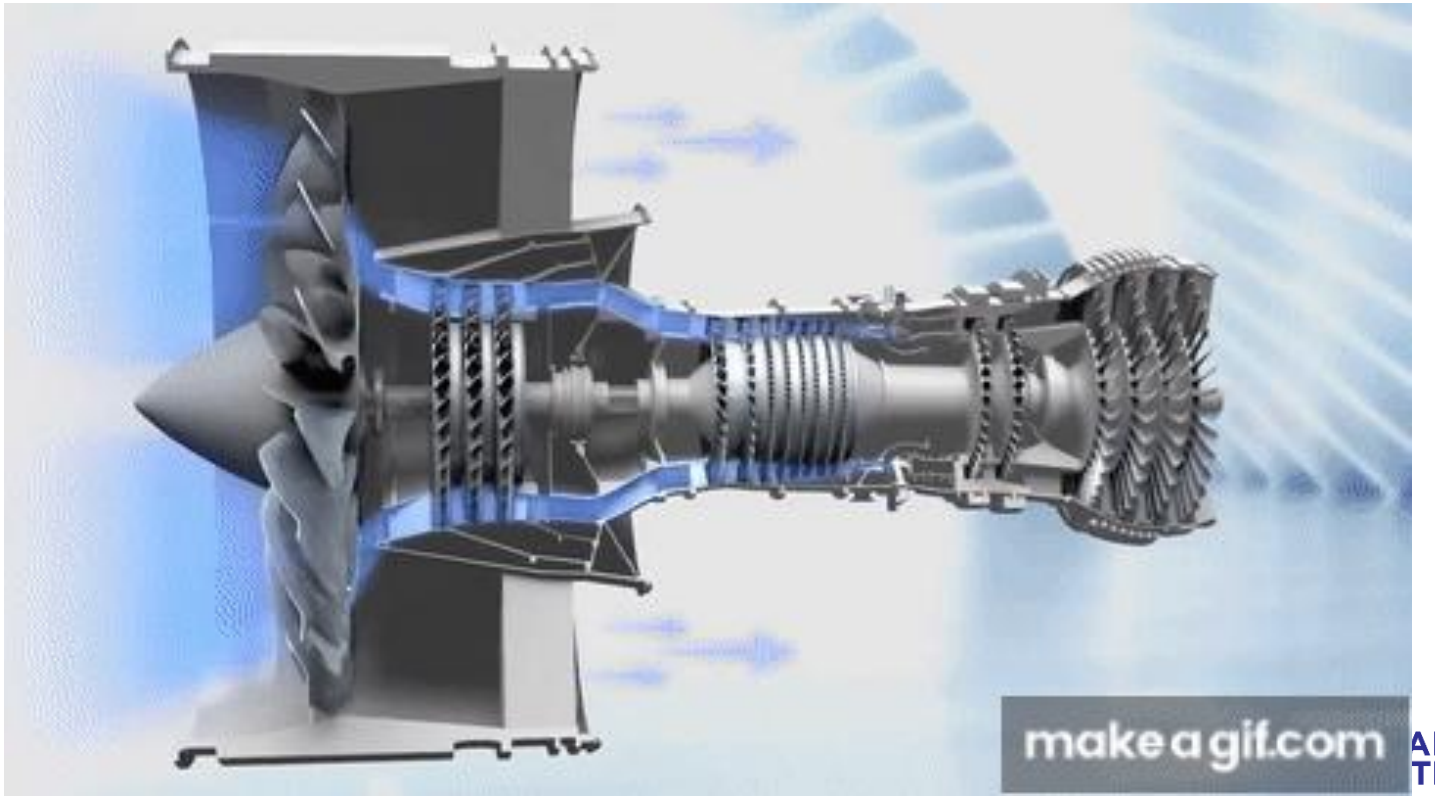
3. Radiation: Heat transfer through electromagnetic waves, without needing a medium



Laws of Thermodynamics

The **Laws of Thermodynamics** are fundamental principles that describe *how energy is transferred and transformed in physical systems.* They are essential for understanding how energy works in everything from engines to biological systems.

Thermodynamics is a branch of physics that deals with heat, work, and temperature, and their relation to energy. Is only concerned to large scale industries.



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Zeroth Law of Thermodynamics

If two systems are each in thermal equilibrium with a third system, then they are in thermal equilibrium with each other.

This law establishes the concept of temperature. It means that if system A is at the same temperature as system C, and system B is also at the same temperature as system C, then A and B must be at the same temperature.

Example: If you place a thermometer in a cup of hot water, the thermometer will reach the same temperature as the water, indicating thermal equilibrium.



First Law of Thermodynamics

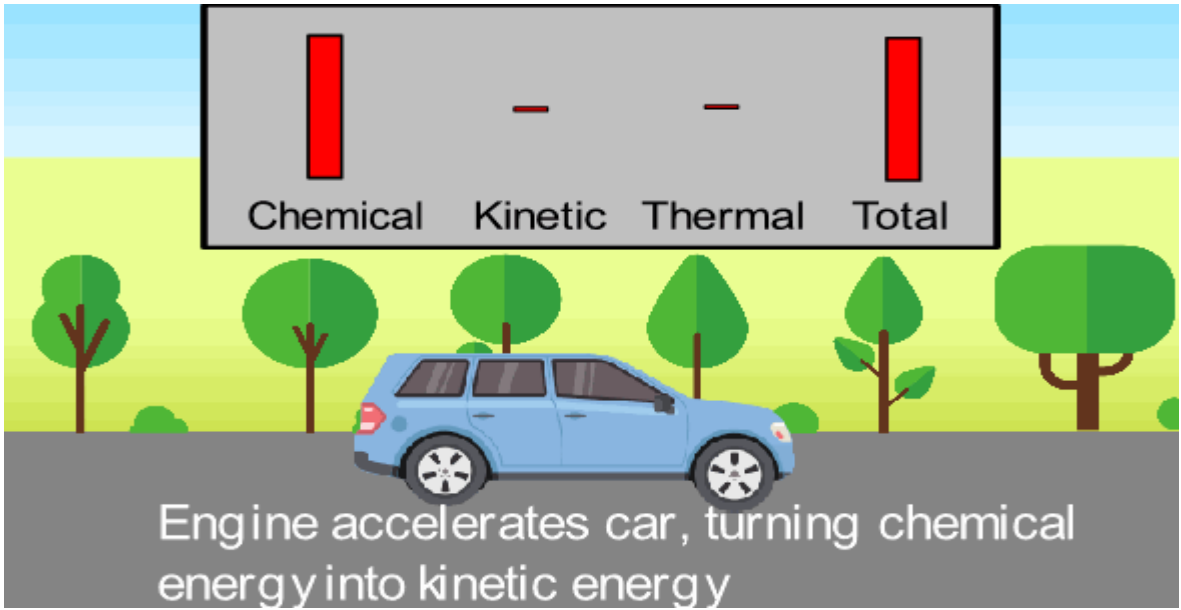
Energy cannot be created or destroyed, only transferred or converted from one form to another.

This law implies that the total energy of an isolated system remains constant.
Energy can change forms, such as from heat to work, but the total amount of energy remains the same.

Example: In an engine, chemical energy from fuel is converted into mechanical work, with some energy lost as heat.

Formula: $\Delta U = Q - W$

- ΔU: Change in internal energy of the system
- Q: Heat added to the system
- W: Work done by the system

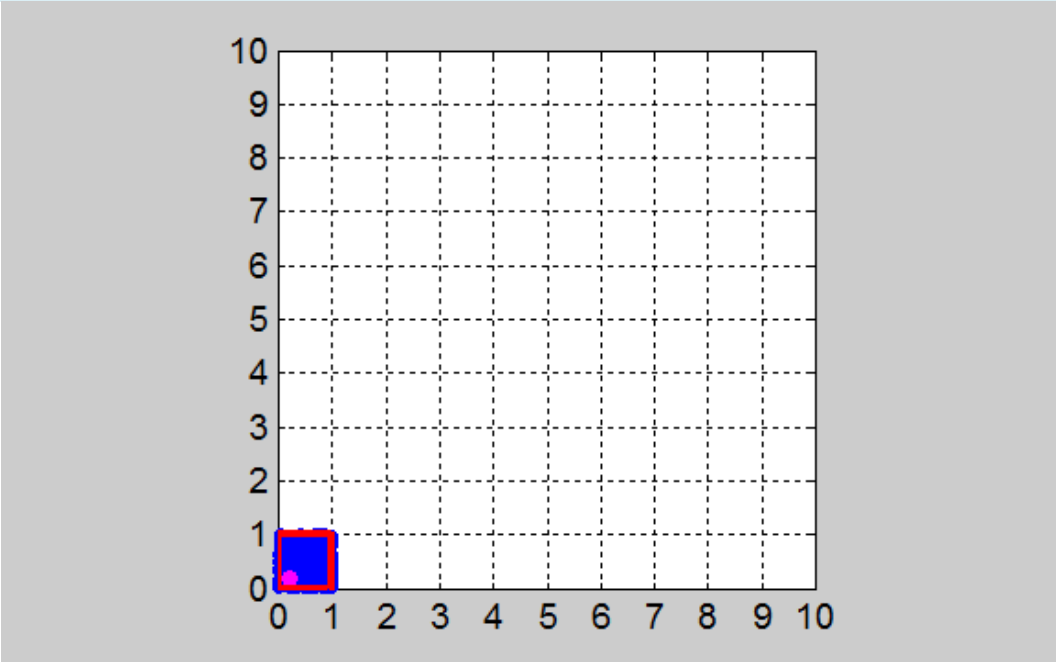


Second Law of Thermodynamics

In any energy transfer or transformation, the total entropy (disorder) of an isolated system always increases over time.

This law implies that natural processes tend to move toward a state of greater disorder or randomness. It also means that energy conversions are never 100% efficient; some energy is always lost as heat.

Example: When you mix hot and cold water, the result is lukewarm water, where the energy spreads out evenly. The process cannot spontaneously reverse, which means entropy increases.



Third Law of Thermodynamics

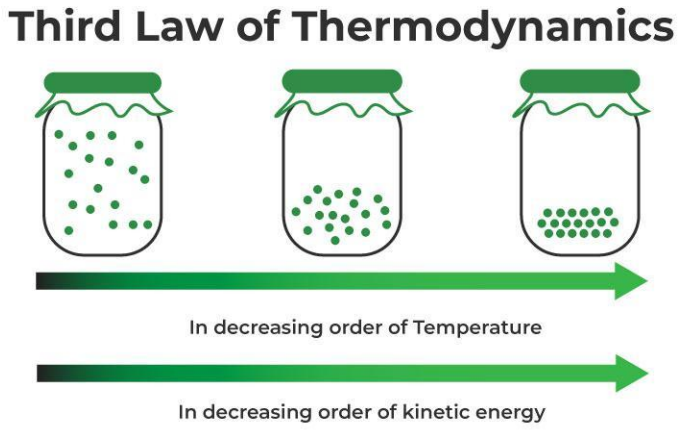
As the temperature of a system approaches absolute zero, the entropy of the system approaches a minimum value, which could be zero for a perfectly ordered system.

At absolute zero (0 Kelvin), a system's particles would be in their lowest possible energy state, and the system would have minimal entropy. However, reaching absolute zero is theoretically impossible.

Example: Certain materials exhibit superconductivity (zero electrical resistance) at very low temperatures, close to absolute zero. The entropy of the system drops significantly, but it never becomes exactly zero.

•In reality, reaching **absolute zero is impossible** because it would require removing every last bit of energy from the system, **which is not feasible. Even the coldest temperatures achieved in labs are slightly above absolute zero.**

This law explains why achieving absolute zero is practically impossible and why materials behave differently as they approach these extremely low temperatures.



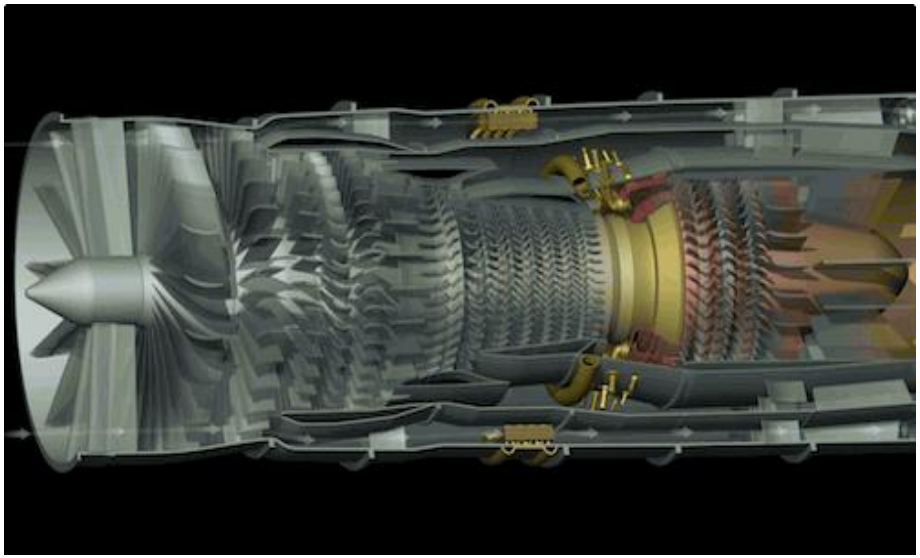
Summary of Laws of Thermodynamics

1. Zeroth Law: Defines temperature and thermal equilibrium.

2. First Law: Energy is conserved; it can neither be created nor destroyed.

3. Second Law: Entropy, or disorder, always increases; energy transfers are never fully efficient.

3. Third Law: As temperature approaches absolute zero, entropy approaches a minimum.





Steam Formation

- **1 Solid ------- ICE**
- **2 Liquid ------ WATER**
- **3 Vapour------ STEAM**
- **STEAM:**

Vapour form of water is called STEAM

Steam is produced by heating water to its saturation temperature.

Types(Staes) of steam

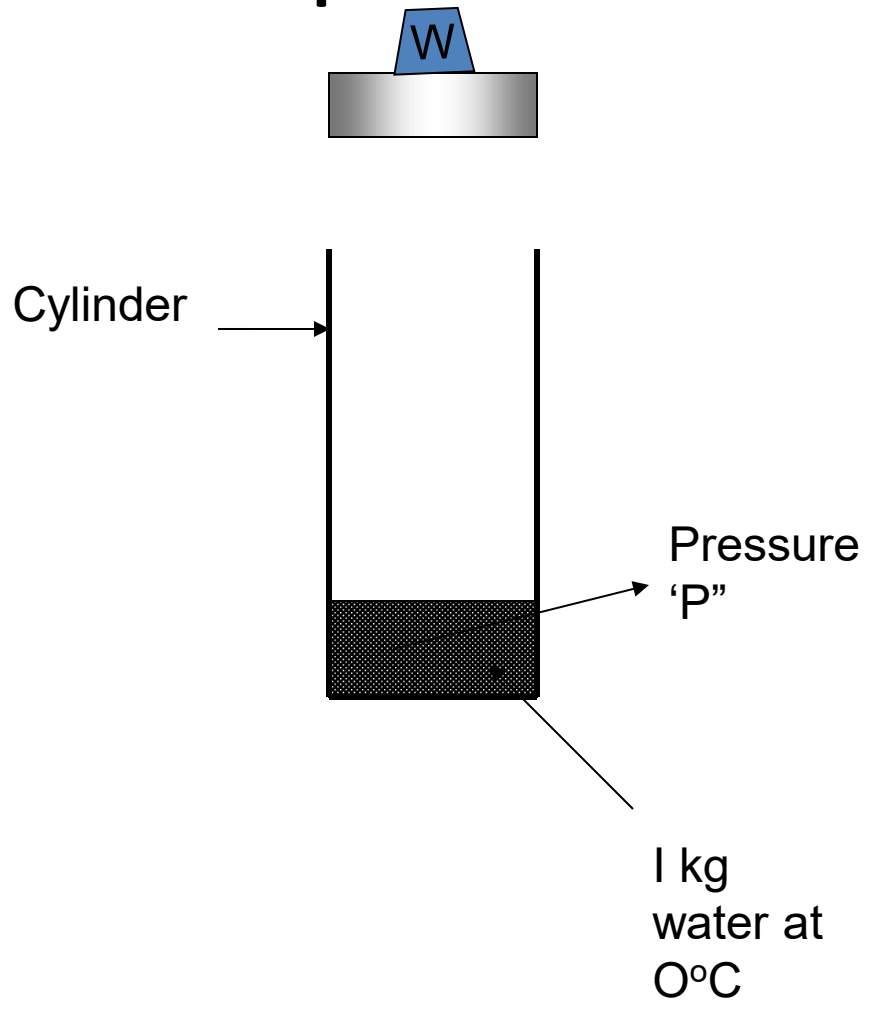
- **Wet Steam :** Steam containing fine water particles in equilibrium is called wet steam.

- **Dry steam :** Steam produced on complete transformation of water into gaseous state at the saturation (boiling) temperature is called dry steam. The saturation temperature depends on pressure.

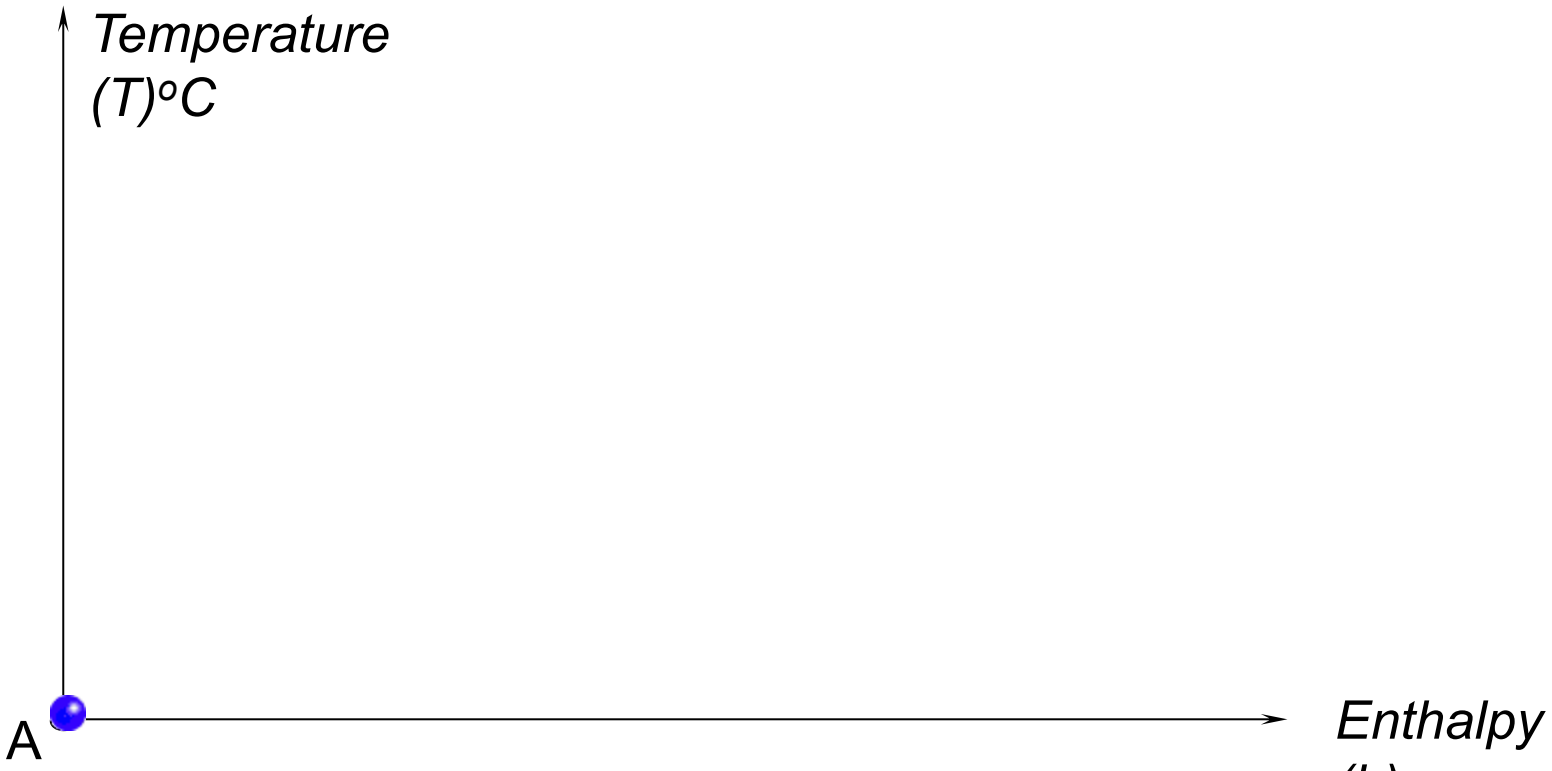
- **Super heated steam:** Steam available at a temperature above the saturation temperature is called superheated steam.

Formation of steam experiment

- Consider 1 kg of water at 0°C taken in a cylinder fitted with a freely moving frictionless piston as shown in fig.



The initial condition of water at 0°C is represented by the point *A* on the temperature-enthalpy graph



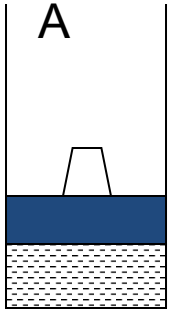


Fig. 1

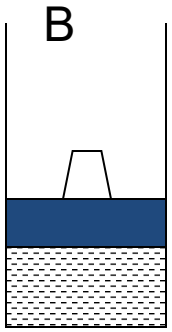


Fig. 2

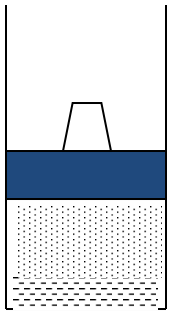


Fig. 3

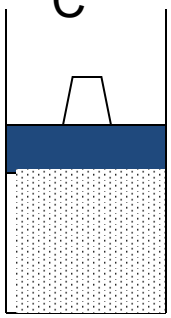


Fig. 4

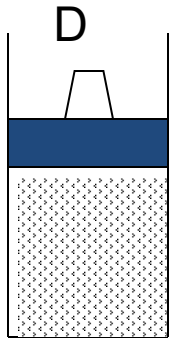
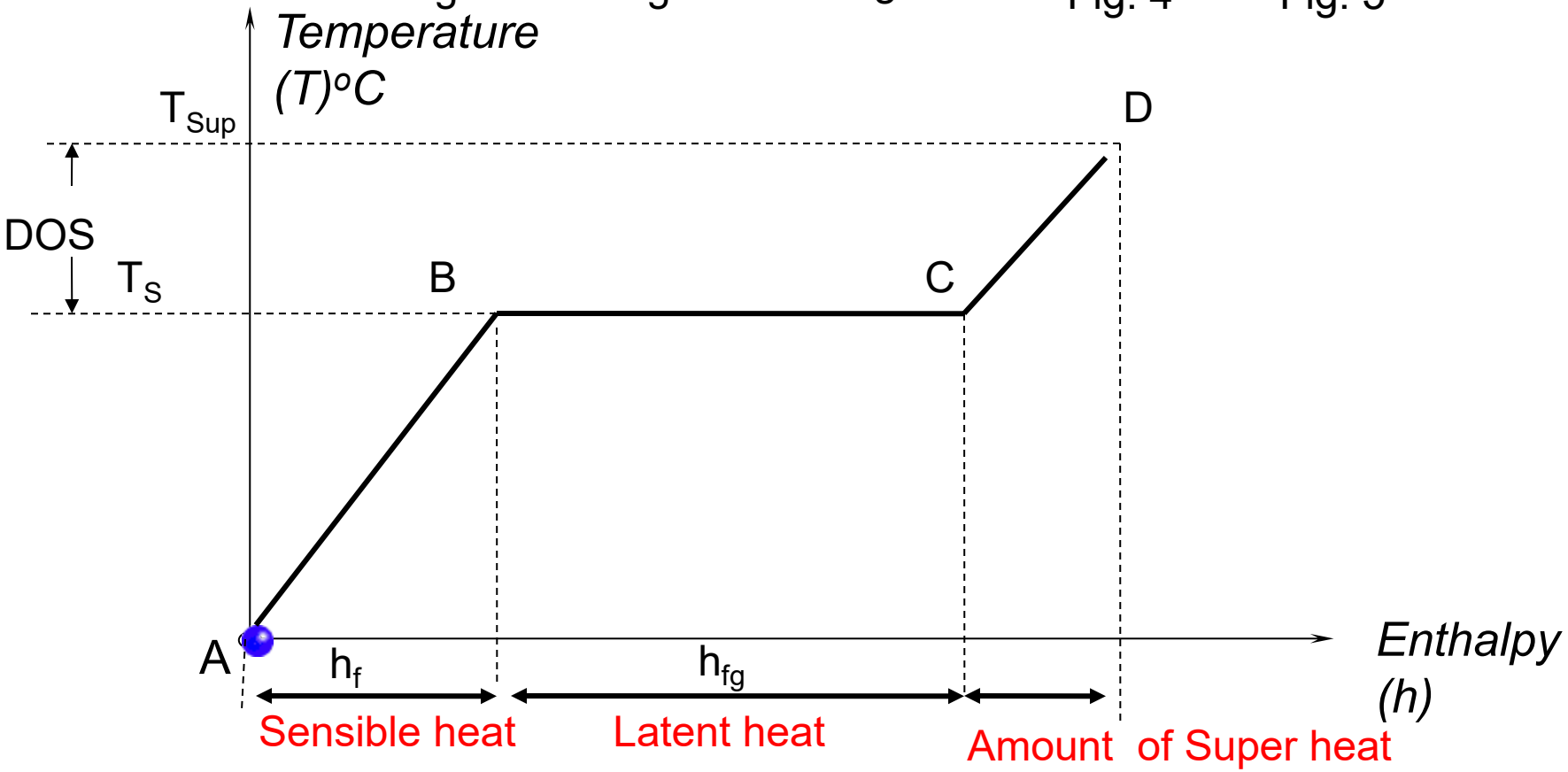


Fig. 5



Measure of Heat contained in steam

- **Enthalpy:**

The heat absorbed by 1 kg of water at 0^0C to transform into steam (wet, dry or superheated) at constant pressure is called enthalpy of steam. It is also known as total heat of steam and expressed in Kilo joules / Kg (kJ/kg)

Definitions

- ***Sensible heat*** **(h_f)** ***:***

*It is defined as the amount of heat required to raise the temperature of **1 kg** of water from **0^0C** to the saturation temperature **T_{sat}** °C at a given **constant pressure.***

- ***Saturation temperature(T_{sat}):***

It is defined as the temperature at which the water transforms into vapour at the stated pressure.

- ***Latent heat of evaporation (h_{fg}):***

*It is defined as the **amount of heat** required to transform **1 kg of water** at saturation temperature into **1 kg of dry steam** at the same saturation temperature at given constant pressure.*

- *The temperature to which steam is heated above the **saturation temperature** at a given pressure is called **superheated temperature (T_{sup}).***

- ***Amount of superheat (AOS) or enthalpy of superheat:***

  It is defined as the amount of heat required to increase the temperature of dry steam from its saturation temperature to any desired higher temperature at the given constant pressure.

- ***Degree of superheat (DOS):***

  It is the difference between the superheated temperature and the saturation temperature at a given constant pressure.

Applications of steam

- Food processing industries.
- Cooking: hotels, restaurants etc.
- Used as a working fluid in steam engines and steam turbines.
- Used in process industries. (Paper Mill)
- Petrochemical industries.
- Washing/drying/sterilizing in hospitals.
- Health clinic / gym.

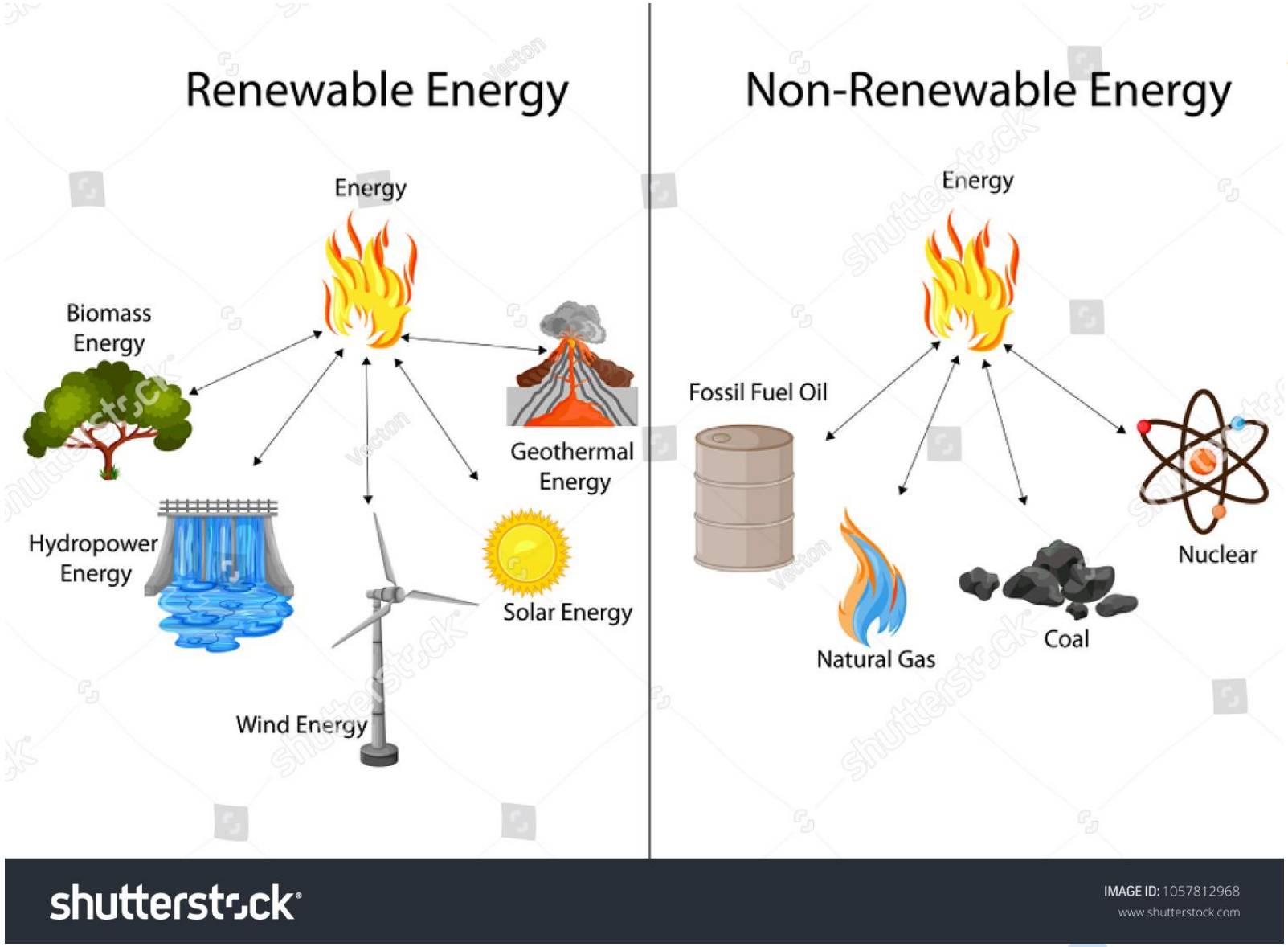
Thank you



Energy Sources & Power Plants







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Non-Renewable

Advantage:

- Available in highly concentrated form.
- Easy to store and transport.
- Reliable supply and they have high energy density.

Disadvantage:

- Highly polluting.
- High running cost.
- A fossil fuel contribute to global warming.
- Its residual products are generally non-biodegradable.
- It produces greenhouse gases.
- Responsible for acid rain.

Renewable Energy Sources

Advantages and Disadvantages

GREENMATCH



Advantages	Disadvantages
Easy regeneration	Highly dependent on weather
Rise in economic growth	Higher installation cost
Abundantly available	High noise caused (wind energy)
Environmentally supportive	Fluctuation in current (PV)
Low maintenance cost	Discontinuous in supply (wind)

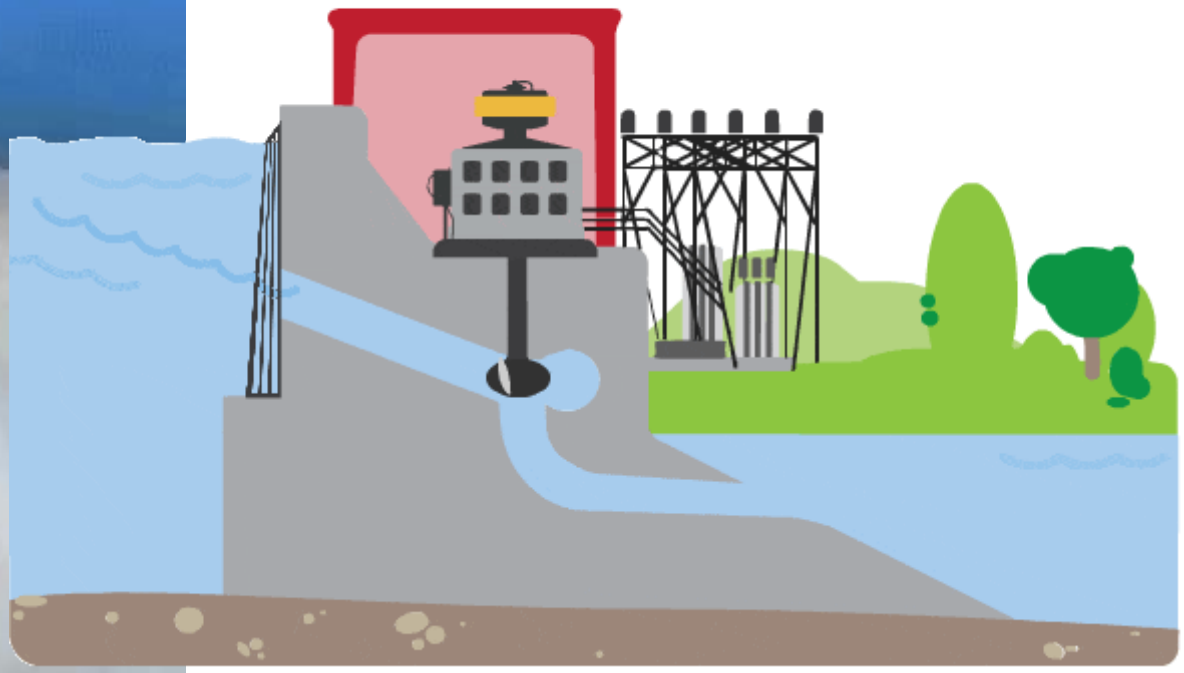
Power Plants

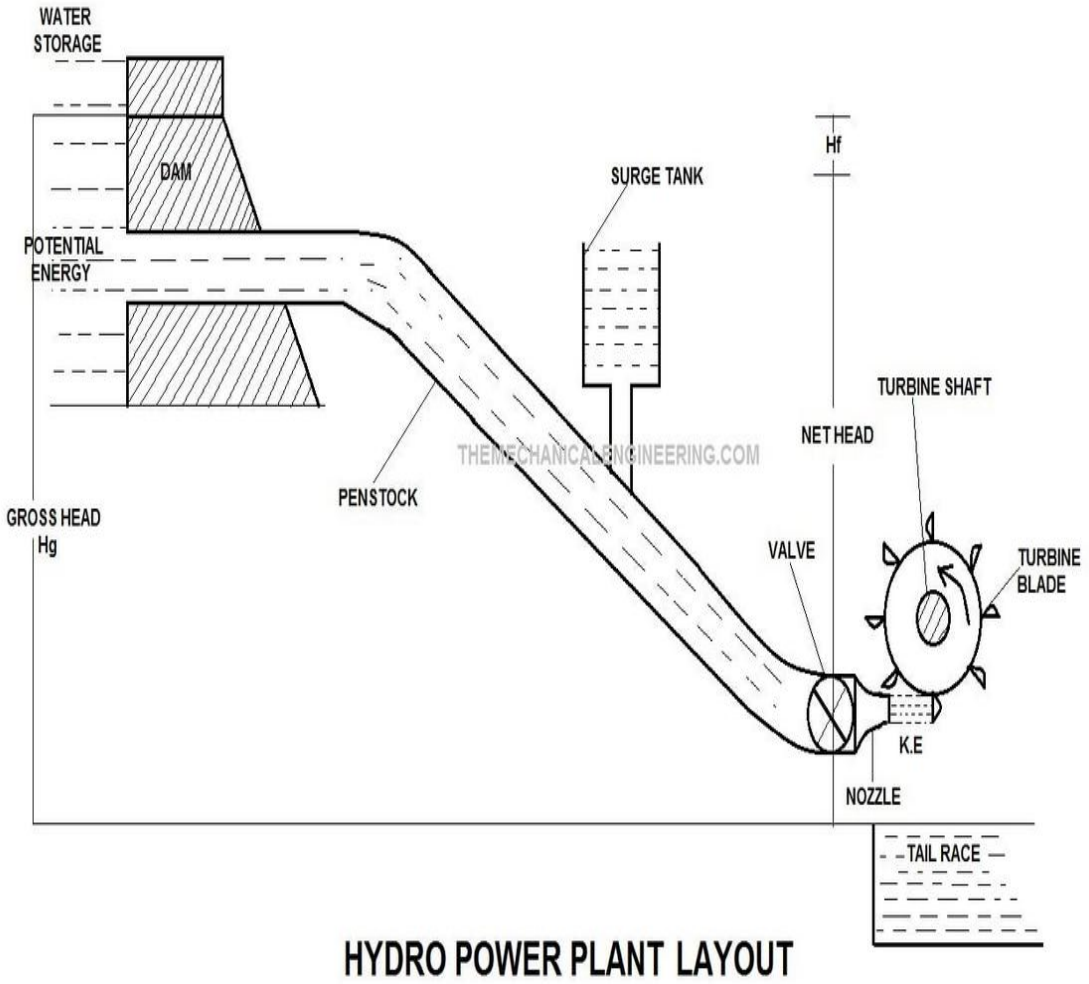


Hydro power

Energy in water can be harnessed and used. Since water is about 800 times denser than air, even a slow flowing stream of water, or moderate sea swell, can yield considerable amounts of energy







HYDRO POWER PLANT LAYOUT

1. Principle of Operation

A hydel power plant works on the **conversion of potential energy of stored water into mechanical energy, and then into electrical energy**. It relies on the fact that water at a height has gravitational potential energy. When it flows down, this energy is converted into kinetic energy, which drives turbines to produce electricity.

2. Basic Working Steps

Step 1 – Water Storage (Potential Energy)

•Water is stored in a **reservoir** at a height using a **dam**.

•The stored water has **potential energy** due to its elevation.

Step 2 – Water Flow to Turbine (Kinetic Energy)

•Water is released from the reservoir through **penstocks** (large pipes).

•As it flows down under gravity, its potential energy changes into **kinetic energy**.

Step 3 – Turbine Rotation (Mechanical Energy)

•The high-speed water jet strikes the **turbine blades**, causing them to rotate.

•The turbine converts the water's kinetic energy into **mechanical rotational energy**.

Step 4 – Electricity Generation (Electrical Energy)

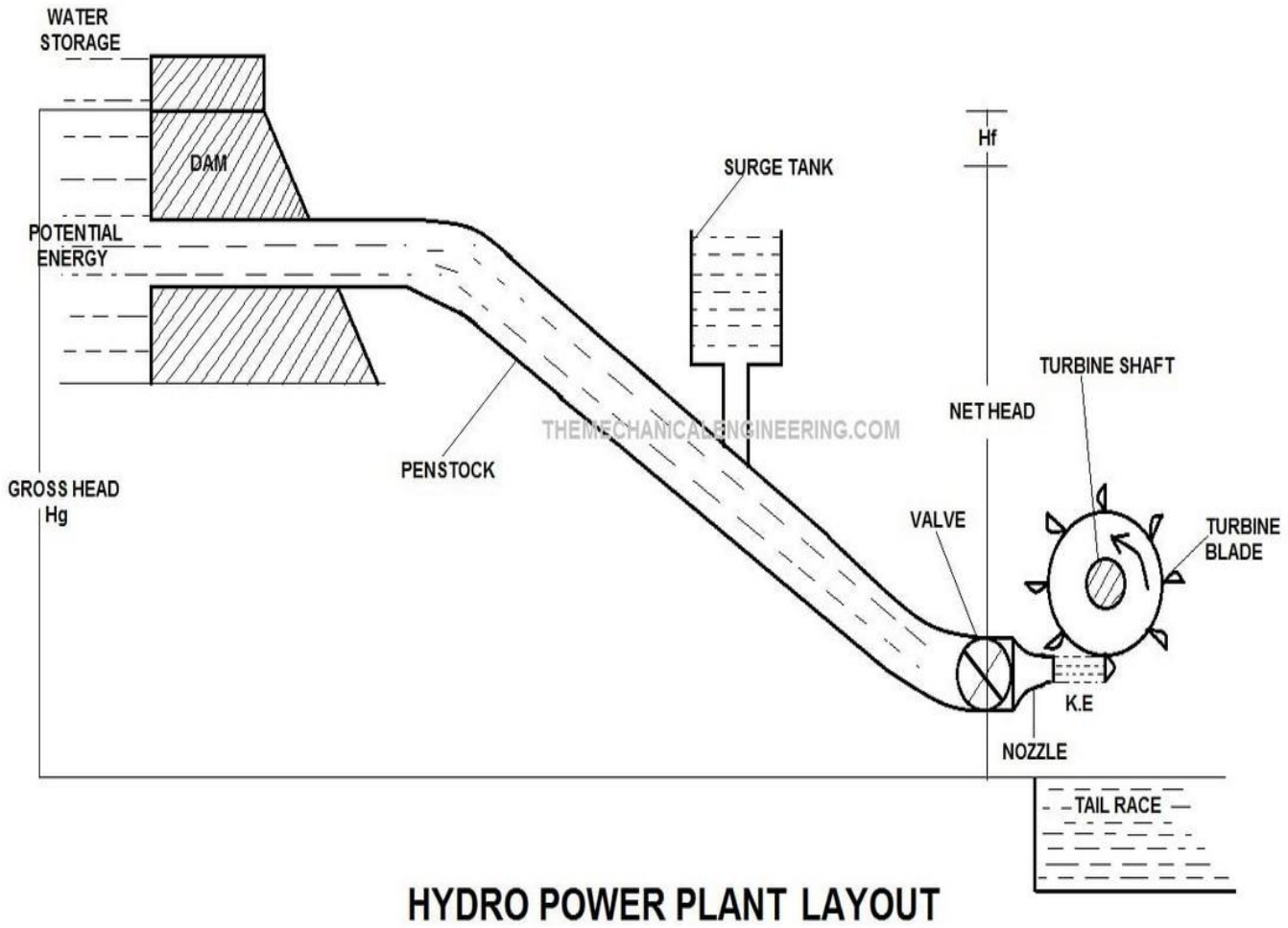
•The turbine shaft is connected to an **electric generator**.

•As the turbine rotates, the generator's rotor spins inside a magnetic field, inducing an **electric current** via **electromagnetic induction**.

Step 5 – Power Transmission

•Generated electricity is stepped up using a **transformer** and transmitted through **power lines** to consumers.


THERMAL POWER PLANT
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HYDRO POWER PLANT LAYOUT

Key Components

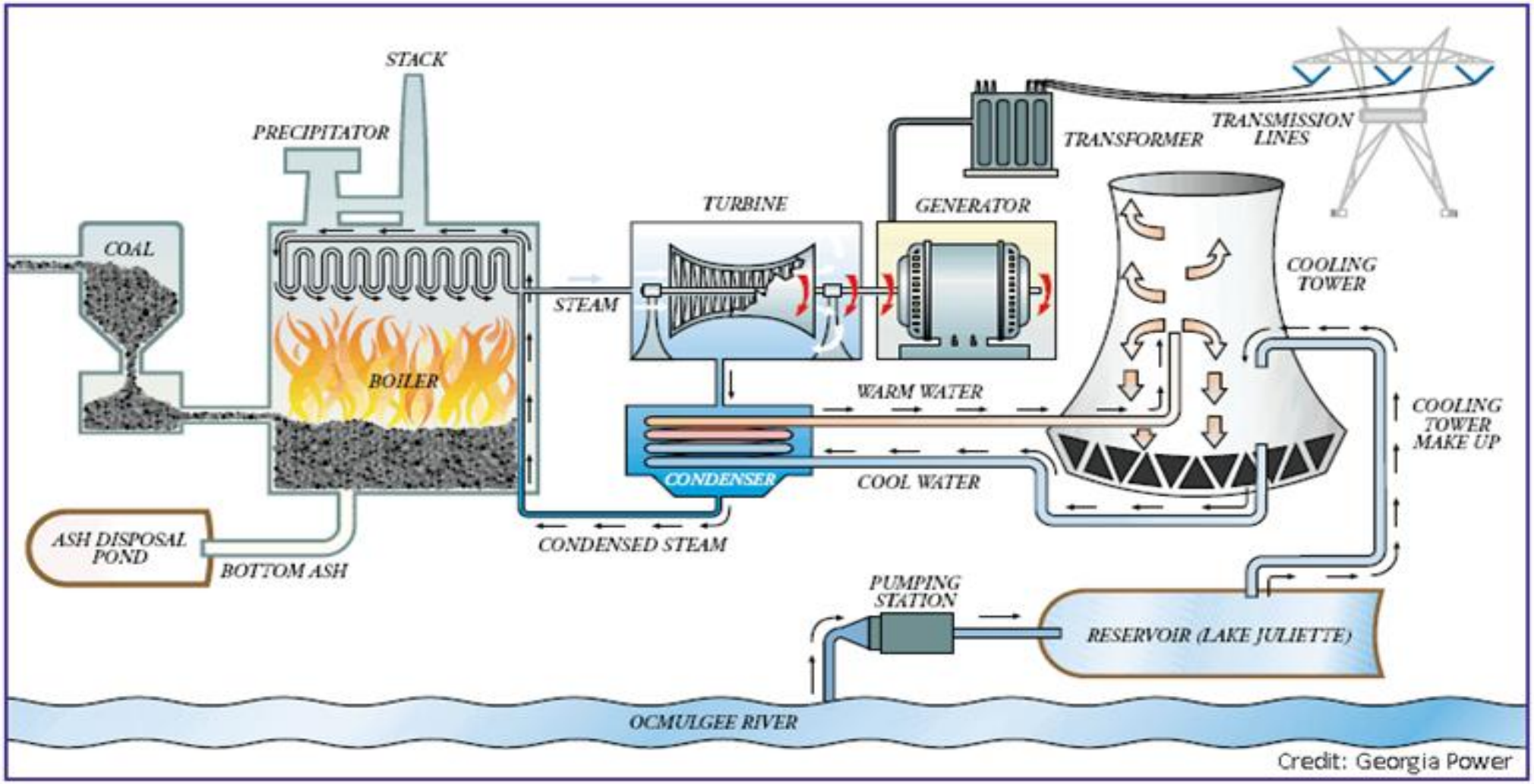
- **Dam/Reservoir** – Stores water at a height.
- **Penstock** – Directs water from reservoir to turbine.
- **Turbine** – Converts water's kinetic energy into mechanical energy.
- **Generator** – Converts mechanical energy into electrical energy.
- **Transformer** – Increases voltage for efficient transmission.

Energy Conversion Summary

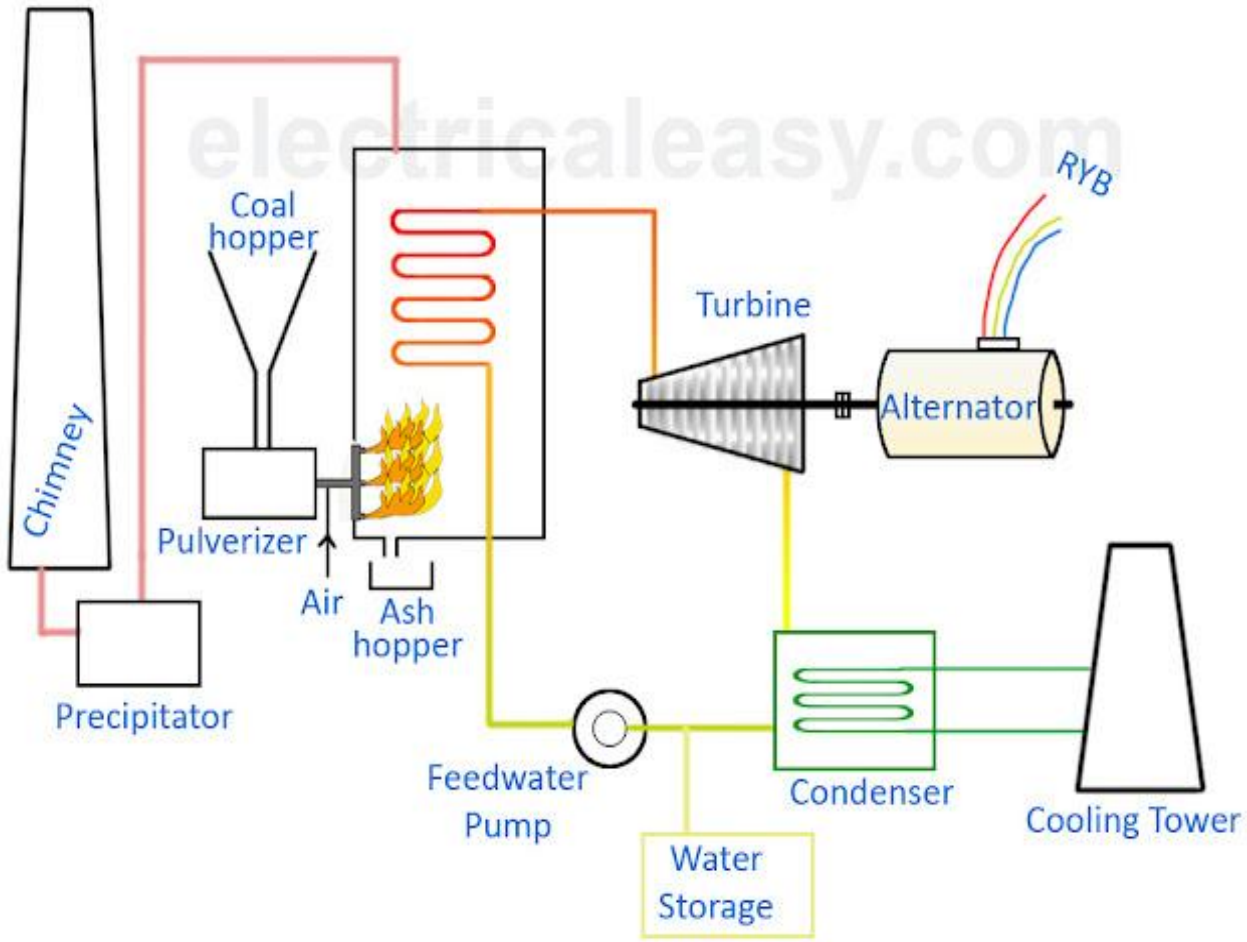
$$\text{Potential Energy (Water at height)} \rightarrow \text{Kinetic Energy (Flowing water)} \rightarrow \text{Mechanical Energy (Turbine)} \rightarrow \text{Electrical Energy (Generato}$$

Thermal power plant

A thermal power plant works on the Rankine cycle.
It converts the heat energy from burning fuels (coal, oil, natural gas, or biomass) into mechanical energy by producing high-pressure steam, which then drives a turbine connected to a generator to produce electricity.



Thermal power plant (Schematic)



2. Basic Working Steps

Step 1 – Fuel Combustion (Heat Generation)

- **Fuel** (e.g., coal) is burned in the **boiler furnace**.
- The combustion releases **heat energy**.

Step 2 – Steam Generation

- Heat from the furnace converts water in **boiler tubes** into **high-pressure, high-temperature steam**.

Step 3 – Turbine Rotation

- Steam is directed onto the **turbine blades** at high speed.
- The **kinetic energy** and **pressure energy** of steam rotate the turbine shaft (mechanical energy).

Step 4 – Electricity Generation

- The turbine shaft is connected to an **electric generator**.
- As the generator's rotor spins in a magnetic field, it produces electricity through **electromagnetic induction**.

Step 5 – Steam Condensation

- After passing through the turbine, the steam enters a **condenser**.
- It is cooled (usually with water from a cooling tower or river) and turns back into liquid water.

Step 6 – Water Recirculation

- The condensed water is pumped back to the boiler for reuse, completing the cycle.

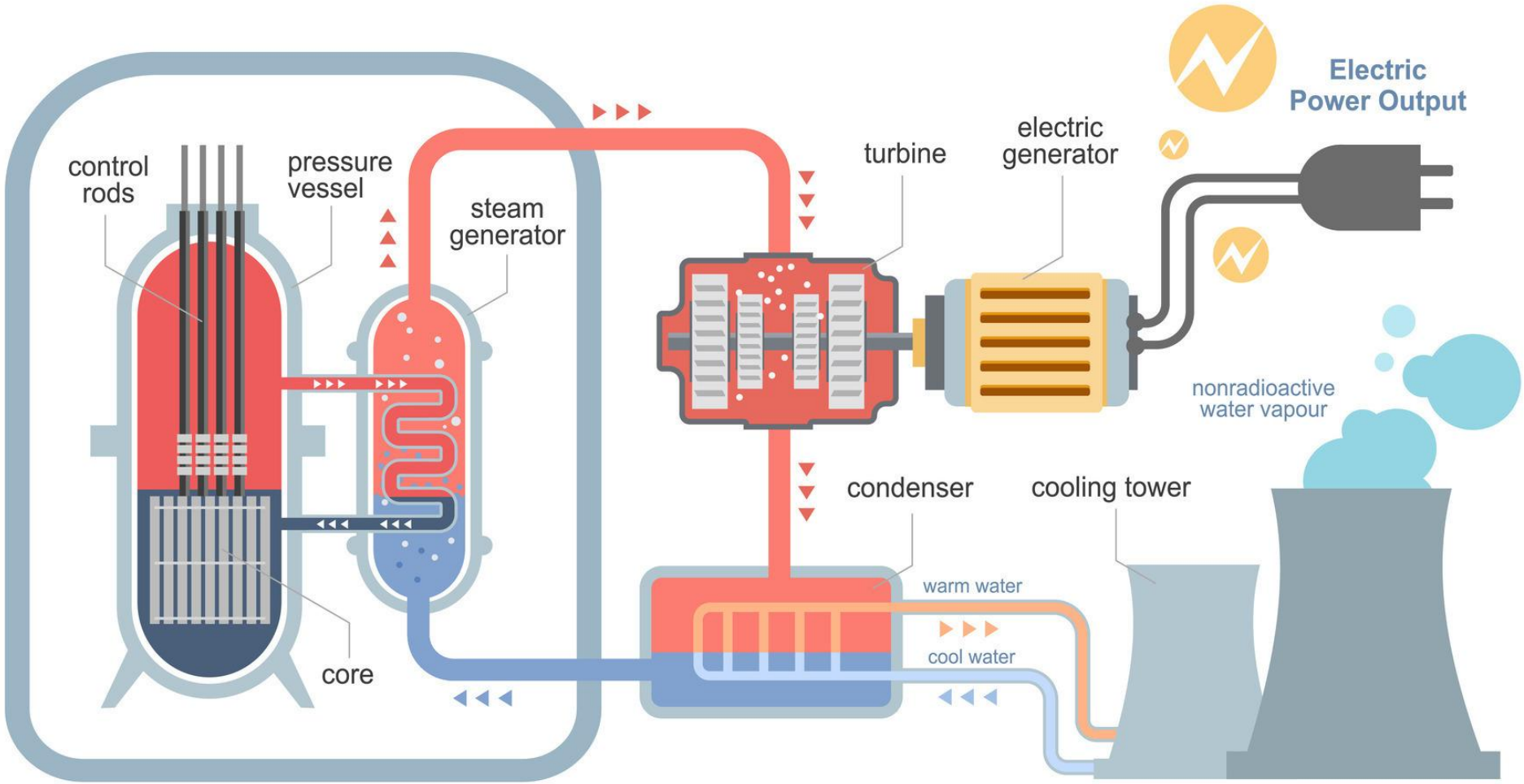
3. Key Components

- **Boiler** – Burns fuel to produce steam.
- **Steam Turbine** – Converts steam energy into mechanical rotation.
- **Generator** – Converts mechanical energy into electrical energy.
- **Condenser** – Cools and condenses steam into water.
- **Cooling Tower** – Removes waste heat from the cooling water.
- **Feedwater Pump** – Recirculates water back to the boiler.

4. Energy Conversion Summary

$$\text{Chemical Energy (Fuel)} \rightarrow \text{Heat Energy (Steam)} \rightarrow \text{Mechanical Energy (Turbine)} \rightarrow \text{Electrical Energy (Generator)}$$

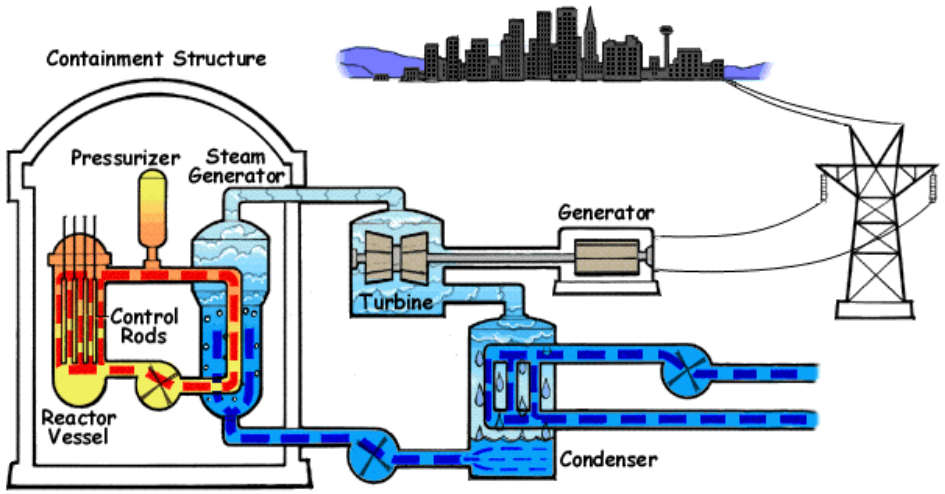
Nuclear power plant



Nuclear Reactor:

•A nuclear power plant works on the **same basic principle as a thermal power plant** — using steam to drive turbines — but instead of burning fossil fuels, it uses **nuclear fission** (splitting of heavy atomic nuclei) to produce heat.

•The most commonly used fuel is **Uranium-235** or **Plutonium-239**.



2. Basic Working Steps

Step 1 – Nuclear Fission (Heat Generation)

- In the **reactor core**, uranium or plutonium atoms are bombarded with neutrons.
- The atoms split (nuclear fission), releasing a large amount of **heat energy** along with more neutrons.
- The chain reaction is controlled by **control rods** (made of materials like boron or cadmium) to maintain safe operation.

Step 2 – Heat Transfer to Coolant

- The heat produced in the reactor is absorbed by a **coolant** (often water, sometimes liquid sodium or gas).
- In most designs, the heated coolant turns water into **high-pressure steam** (either directly or via a heat exchanger).

Step 3 – Turbine Rotation

- The high-pressure steam flows to the **steam turbine**, making it rotate and converting **thermal energy** into **mechanical energy**.

Step 4 – Electricity Generation

- The turbine shaft is connected to an **electric generator**.
- The generator's rotor spins inside a magnetic field, producing electricity by **electromagnetic induction**.

Step 5 – Steam Condensation

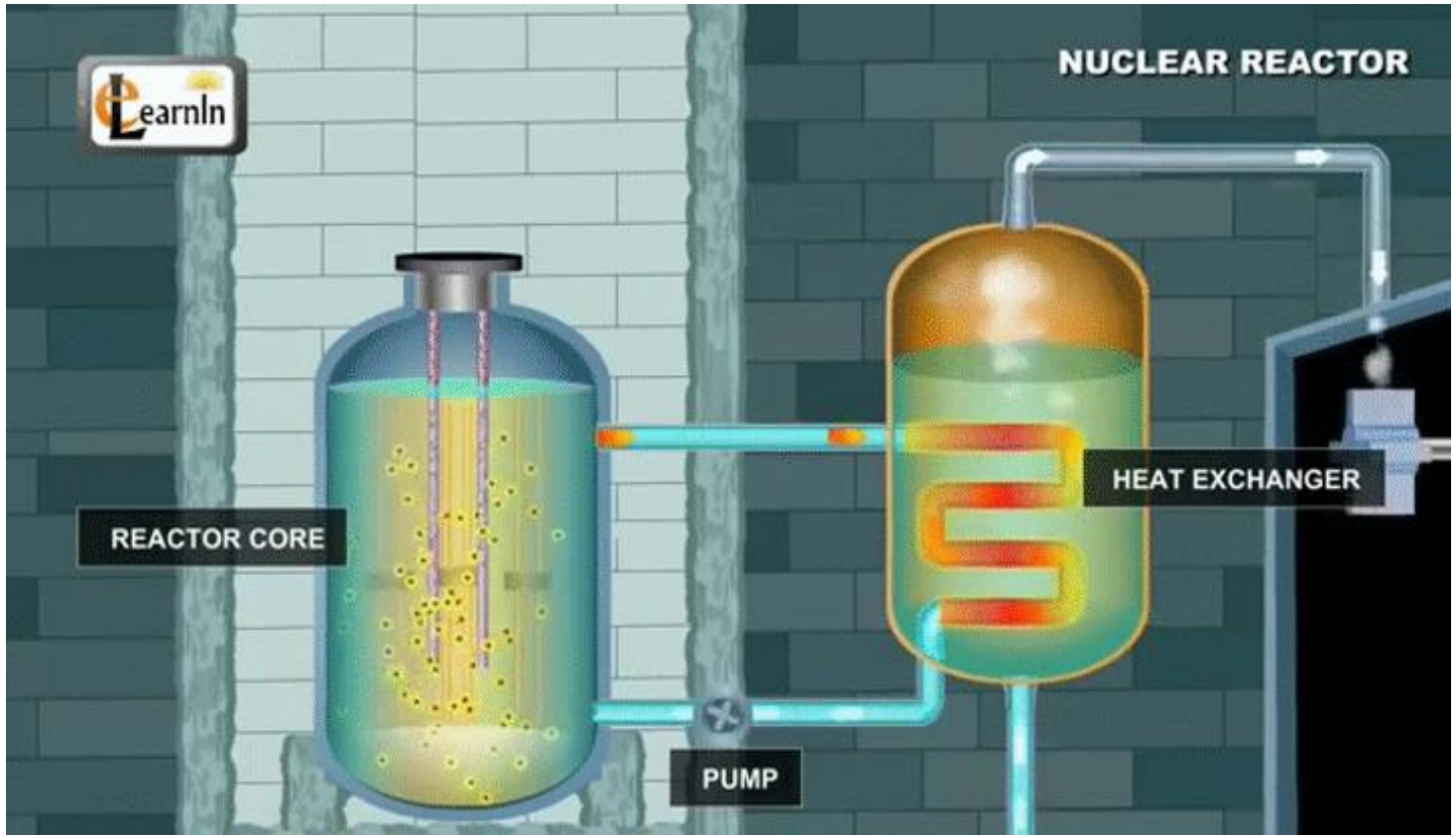
- After passing through the turbine, steam is cooled in a **condenser**, turning it back into water.
- Cooling is usually done using water from cooling towers, rivers, or seas.

Step 6 – Recirculation

- The condensed water is pumped back into the h exchanger or reactor to repeat the cycle.

Nuclear power plant

•When these atoms split, they release a huge amount of heat.



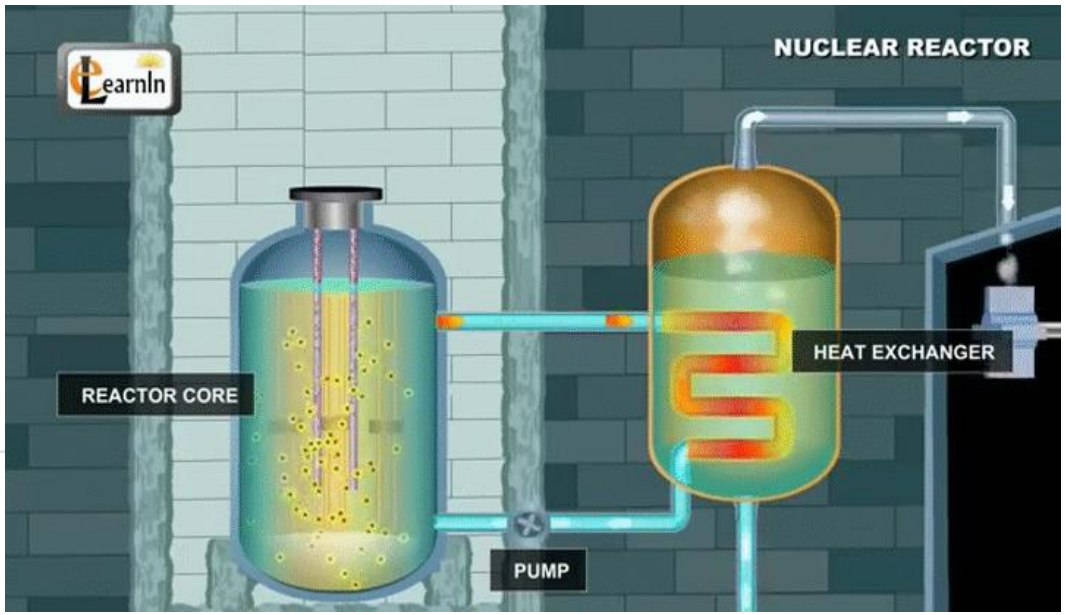
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•The most commonly used fuel is **Uranium-235** or **Plutonium-239**.

Key Components

- **Nuclear Reactor** – Produces heat via fission.
- **Control Rods** – Regulate the chain reaction.
- **Moderator** – Slows down neutrons to sustain the reaction (e.g., graphite or heavy water).
- **Coolant** – Transfers heat from the reactor core.
- **Steam Turbine** – Converts steam energy into mechanical rotation.
- **Generator** – Produces electricity.
- **Condenser** – Converts steam back into water.
- **Cooling Tower** – Removes waste heat.



Energy Conversion Summary

$$\text{Nuclear Energy (Fission)} \rightarrow \text{Heat Energy (Coolant)} \rightarrow \text{Mechanical Energy (Turbine)} \rightarrow \text{Electrical Energy (Generator)}$$